main.pdf

PDF

what are your general thoughts about this thesis draft?

Checking the disclosures and citations against the body

This is a strong project with real problems in how the evidence is presented. I only read the text (not the figures), so these notes come from that.

What's good. The problem is real and well chosen. The Magtajas/RA 7160 framing is solid, the data access through the Sangguniang Panlungsod is a genuine asset, and the retrieve-then-entail design is well justified for LGU hardware. The cost-sensitive threshold is a thoughtful touch, and so is the VLM error analysis on degraded scans. Being open about the incomplete annotation and the weak Tier 2 numbers is to your credit. The weak spot is that the abstract and conclusions claim more than your results support.

What I'd fix first

1. **Headline claims versus your own tables.**
   - The "100% Recall@50 over 27,357 enactments" claim doesn't match Chapter 4. Table 4.7, the only full-corpus test, covers BM25 variants only, and its BM25 Recall@50 is 43.14% at about 330 ms. Table 4.9 shows BM25 at 99.7% and 1.39 ms, so it must use a much smaller pool. The dense and hybrid methods are never run on the full corpus. In Table 4.9, plain MiniLM also beats RRF on MRR (0.7388 vs 0.7040) and Recall@10.
   - Tier 2 reports 28.57% section-level sensitivity, 25% precision and 50% accuracy. "100% document triage recall" is trivial when all six drafts contain planted defects, so "ready for institutional deployment" and "comprehensive empirical proof" aren't defensible.
   - The 5.79× speedup (73.47 s → 12.68 s) never appears in a results table. Table 4.18 shows a 28.15 s baseline, about 2.2×.
   - Latency is reported at k=15, but the design and the "under 30 seconds" claim assume k=50. At about 121 ms per pair, 500 pairs is about 60 s.
2. **The benchmark.**
   - The 350 pairs use only about two or three premises per domain, around two dozen in total.
   - Many items are exact duplicates (GT-001/004, GT-002/005, GT-016/019), so random splits and 5-fold CV leak.
   - Entailment hypotheses often say "in strict observance of Section X of RA Y", which makes them easy to spot. Neutral items are random unrelated topics rather than hard topical negatives.
   - Splits should be grouped by premise, and duplicates removed.
   - The labels are the proponents' own, with hypotheses drafted by an LLM and adviser verification. The 15-researcher process and Fleiss' kappa are described in Chapter 3 as if they happened, but Chapter 4 says they didn't. Make Chapter 3 consistent with what was actually done.
   - Section 3.5.3 says Gemini drafted the hypotheses, but the AI disclosure (Appendix O) lists only code, LaTeX and proofreading. Fix that before the defense. The disclosure also says 10 draft ordinances where the body says 6.
3. **Statistics.**
   - Section 2.10.5 argues against k-fold CV on fewer than 1,000 items, then the headline 88.30% is a 5-fold CV mean.
   - Several "unseen test" accuracies in Table 4.13 (86.54%, 82.69%, 84.62%, 80.77%) are impossible on N=53 but equal 45/52, 43/52, 44/52 and 42/52. They look like validation numbers.
   - McNemar p=0.0388 fails your own Bonferroni threshold of 0.0167, yet the abstract says "significantly outperforming".
   - The b/c counts in Table 4.15 don't reconcile with the reported accuracies.
   - Table 4.14 has identical rows across domains and perfect 1–6 ranks everywhere. The Friedman χ^2 is 18.42 in one place and 40.00 in another, and the zero-shot McNemar χ^2 is 7.14 vs 19.31.
   - Table 4.12 shows early stopping at epoch 5 with patience 4, which wouldn't trigger.
   - The balanced benchmark has about 33% conflicts but real drafts have under 1%, so τ^*

and precision won't transfer. Tier 2 hints at this.

4. **Inconsistencies a panelist will catch.**
   - τ* is 0.45, 0.42 or 0.35 depending on where you look.
   - Class counts are 117/117/116 in 3.5.4, 112/128/110 in Table L.1 and 130/110/110 in Table N.3.
   - Local operative provisions are 9,382 vs 9,487, and the local topic split sums to 1,664, not 1,925.
   - The RRF hierarchy boost is 1.20 vs 1.10, and BM25 k1 is 1.5 vs 1.2.
   - Per-pair latency for the same model is 14.8 ms vs 120.9 ms.
   - The 2.15% CER is attributed to Gemini in Table 4.2 but to Document AI in 4.4.4. The Gemini-assisted gold reference also makes that comparison circular.
   - Three of the eight "Supreme Court" Davao cases (mining, smoking, speed limit) have no cited ruling. The model's "88%" appears for five of seven contradictions, which looks odd.
   - The codebook in 2.10.2 says an added local fee is Neutral, but GT-238 labels a local firearm permit with a fee as Contradiction.
   - The symbolic ceiling verifier in the ablation is never described in the methodology.
5. **Scope gaps.** Horizontal (local-vs-local) consistency is a stated objective, but nothing evaluates it. Host C, your "authentic" edge machine, has no reported results. The topic model's 97% "accuracy" is measured against its own cluster labels. The benchmark's domain labels also don't match the topic-model domains (price control and DRRM sit under "Executive Issuances").

Smaller things.

- Sections 2.8.4 and 2.8.6 repeat three paragraphs verbatim.
- Nearly every table and figure is followed by a formulaic "This table is included to…" paragraph, which could be cut substantially.
- Words like "proves", "decisively" and "extraordinary" should be toned down.
- Table 4.20's caption is garbled LaTeX, and "<" renders as "¡" in Tables 4.17 and 4.18.
- Spot-check citations. [175] (an NIJ criminal justice article), [223] (a text-games paper) and [240] (ANLIzing ANLI, cited for a "70% overhead reduction") look mismatched to what they're cited for. [1] and [33] don't obviously support the annotator-workload claim. Entries 24/25, 89/90 and 205/206 are duplicates.

If you want, I can turn this into a prioritized fix list by chapter, or dig into one area, such as redesigning the benchmark splits or reconciling the numbers.

can you also suggest fixes or how we can adjust and defend them in the paper? like provide some solutions for each bullet

Weighing the horizontal-conflict scope and the topic-model claims ▸

Here are fixes and defense angles for each point, sorted by effort. Some need only rewording, some are re-runs with your existing code, and a few are new work if time allows.

Start here (highest payoff)

1. **Dedupe the benchmark and re-split by premise.** Group by the national statute section so no premise appears in both train and test. Report grouped 5-fold CV. This is a few hours with existing code and fixes the biggest validity threat.
2. **Run Stage 1 on the full 178k-provision corpus for every method** (dense, hybrid and BM25) and report R@k, MRR and latency in one table. Embeddings are precomputed, so this is cheap.
3. **Build a numbers registry.** One script or JSON file generates every reported figure (counts, τ*, latencies, p-values), and LaTeX pulls them in via macros. This kills most of the inconsistencies at once.
4. **Reposition the claims.** Describe the work as a "controlled benchmark plus a feasibility prototype", not "ready for institutional deployment".

1. Headline claims vs. your tables

Recall@50 = 100%

- Find out what pool Table 4.9 used. If it was a restricted pool, label it "controlled-pool recall" and add a full-corpus column. If full-corpus dense/hybrid recall is lower, report it and defend k=50 as the safety buffer.

- With about two dozen unique premises, “350 queries” really means about 24 targets. Report recall per unique premise too.
- On the MiniLM-vs-RRF result, drop “hybrid elevated recall across all tiers”, which your table contradicts. Defend RRF on a different basis: R@50 is what Stage 2 consumes, early-rank differences only affect display order, and hybrid protects against dense failures on exact citations and numeric ceilings. Support that with a per-query-type breakdown (numeric/citation queries vs. paraphrased).
- The ablation row where removing dense embeddings changes nothing (±0.00%) undermines the dense component. Re-run it or explain it.

Tier 2 (28.6% sensitivity)

- Rename it a “pilot stress test” and don’t claim readiness.
- Use your own Eq. 3.19 to split failures into retrieval misses and classification misses. Add an oracle-retrieval run (feed the gold premise) so Stage 2 alone is measured.
- Re-run at k=15 and k=50. The text mentions a top-8 window, which is a third k value.
- Add drafts with no planted violations. “100% triage recall” is trivial when every draft is defective, so report document-level specificity and an “always flag” baseline.
- Defense: it’s an open-corpus test where retrieval, not inference, is the bottleneck, and that is a useful finding for future work.

5.79× speedup

- Either add the 73.47 s measurement to a table, or report the 2.2× from Table 4.18. Run a quick 2×2×2 timing ablation (boilerplate bypass × dynamic padding × batching) so each optimization’s contribution is explicit.

k=15 vs k=50

- Make k=15 the default operating point. Your own tables show R@20 is about 99.7–100%, which justifies it. Offer k=50 as a “deep audit” mode with its measured latency (about 40 s for 7 sections, 60 s or more for 10). Fix the “under 30 seconds” sentence in 3.9.2.
- Label the hardware on every latency figure (14.8 ms vs. 120.9 ms is probably GPU vs. CPU).

2. The benchmark

- **Few premises.** Best fix: add 4–6 premises per domain and write new pairs. Minimum fix: hold out whole premises in the test set and rename the benchmark a “controlled diagnostic benchmark”. Defense: constructed conflicts have unambiguous gold labels (you can cite the statutory mechanism), and Tier 3 plus the court cases test real-world transfer.
- **Duplicates.** Remove exact duplicates, then recompute N, the power analysis and the splits. Say plainly how many unique pairs there are.
- **Citation cues** (“in strict observance of Section X of RA Y”).
  - Run a hypothesis-only baseline and a cite-the-premise regex baseline.
  - Re-evaluate with citation phrases and the templated closing sentences masked. This is cheap and you already cite Gururangan and Poliak for the logic.
  - Add Contradiction items that cite the statute and Entailment items that don’t.
  - If the baselines score high, present the masked score as the primary number.
- **Easy Neutrals.** Add 30–50 hard Neutrals on the same statute or topic but a different obligation, and report them as their own tier.
- **Annotation story.**
  - Chapter 3 should read as “adviser-verified benchmark, with expert validation designed and pending”.
  - Get any real IRR number you can. Options: both proponents and the adviser label a blind 60-pair stratified subsample and report Cohen’s/Fleiss’ κ, or report whatever SP panels did return as a pilot.
  - Defense: labels come from the stated statutory mechanism, not subjective judgment, so noise is low but difficulty is also low. Say that.
- **AI disclosure.** Update the statement and Appendix O to say Gemini was used for syntactic drafting of hypotheses (3.5.3) and as the OCR reference starting point. Fix 10 vs. 6 drafts. Keep the prompts and few-shot templates in the repo. A panelist will ask “did the LLM generate your labels?” and the answer is “no, we set premise, mechanism and label; it only phrased the text.”

3. Statistics

- **CV vs. holdout.** Use one coherent protocol: fixed validation for tuning, and premise-grouped CV for final estimates, because a 53-item test set has about ±9pp CI. Report Wilson or bootstrap intervals. Re-read Vabalas et al. before citing them. I recall their conclusion being less favorable to small holdouts than 2.10.5 implies, so check that the citation supports what you say.

- **Power claim.** The w=0.30 chi-square calculation isn't a classifier-comparison power analysis. Replace it with CI widths or an MDE statement: at N=53, only large gaps are detectable.
- **Impossible N=53 accuracies.** 86.54, 82.69, 84.62 and 80.77% equal 45/52, 43/52, 44/52 and 42/52, so they look like validation numbers. Re-run the final checkpoints once on the real test set and save per-item predictions as JSONL in the repo so numbers can be audited.
- **McNemar vs. Bonferroni.**
  - Run McNemar on out-of-fold predictions over all (deduped) pairs for more power.
  - Pre-specify DeBERTa vs. BERT as the single primary comparison, or use Holm.
  - If it doesn't survive correction, write "p = 0.039 uncorrected; suggestive".
  - Recompute b and c from predictions. Your counts don't match the reported accuracy gaps (12 vs. 3 on 53 items implies a 17pp gap, not 3.85pp).
- **Friedman table.** Identical rows and perfect ranks across all domains look copied. Recompute per-domain macro-F1 from real out-of-fold predictions, use average ranks for ties, and take p from `scipy.stats.friedmanchisquare`. χ^2=40 with df=5 gives p of about 1e-7, not 0.00035, and the text also says 18.42. With only 8 non-independent blocks, a paired bootstrap CI may serve you better.
- **Early stopping (Table 4.12).** With patience 4 and a 5-epoch cap, it can't trigger. Either say "best checkpoint by validation loss (epoch 3)" or set max epochs to 10 and show the real stop. Pick one criterion (Macro F1 in Table 3.8, val loss in 3.8.3). Show the base model's curve, since that's your proposed model.
- *Threshold τ.** Compute F2 from Table N.2: it peaks at 0.40 (about 0.928 vs. 0.925 at 0.45), while F1 peaks at 0.45. If you claim an F2-tuned τ^*, the data says 0.40. Also, Table N.2 appears to be computed over all 350 pairs, which includes training data. Recompute it on validation or out-of-fold predictions and pick one τ^* everywhere.
- **Prevalence shift.** Add a PPV-vs-prevalence table: PPV = sens·p / (sens·p + (1−spec)(1−p)). At 1% prevalence with about 93% sensitivity and 96% specificity, PPV is only about 20%. Defend it as a recall-first triage tool with a user-adjustable threshold, and report AUPRC.

4. Inconsistencies

- Use the numbers registry for τ^*, class counts, provision counts, RRF boost (1.20 vs. 1.10) and BM25 k1 (1.5 vs. 1.2). Report what the code that produced the tables actually used. Appendix D shows 1.10 and 1.2.
- Class counts: generate Tables L.1, L.2 and N.3 from the dataset file, and delete the "117 per class" claim.
- Local topic holdout: 1,331 + 333 = 1,664, not 1,925. Explain the exclusions or fix it.
- Cloud CER attribution: pick Gemini or Document AI, and state that Gemini helped draft the gold reference, which favors it. Also confirm how many documents were human-verified. Table 4.3 shows 4, while 3.4.2.1 describes 60. If it's 4, scale the Table 4.2 claims down.
- Tier 3 cases:
  - Relabel the three without cited rulings (mining, smoking, speed) as "control ordinances with author-analyzed labels", not Supreme Court validation.
  - Reconcile "8 cases" with "11".
  - Reconcile ARC Investors citing §135 in one table and §69(f) in another.
  - Verify each gold label against the actual dispositive portion.
  - Explain the repeated "88%" (calibration binning? a rule override?). Disclose any override.
  - Present these as qualitative illustration, not "100% alignment".
- Codebook: 2.10.2's "added local fee = Neutral" conflicts with GT-238 and with your own 2.2.5 (implied preemption). Revise 2.10.2 to match: a local permit or fee that conditions a nationally licensed activity is a Contradiction, while a fee on a locally regulated activity is not.
- Symbolic ceiling verifier: describe it in 3.7 (regex extraction of peso amounts checked against the ₱5,000 cap, Tier 1 only), or remove it from the ablation.

5. Scope gaps

- **Horizontal consistency.** Fastest: scope it down. Say the architecture supports it but only vertical preemption was evaluated, and edit the objectives and abstract to match. Defense for the omission: local-vs-local differences are often intentional updates, so NLI's Contradiction label isn't the right target (you'd need Duplicate/Supersedes labels). Stronger: build about 30 adviser-verified local-vs-local pairs and evaluate them.
- **Host C.** Run the 14-section draft on the Ryzen 5 4500U and report latency and RAM. It's about an hour of work. Otherwise drop "authentic" and call Host A a throttled proxy.
- **Topic model "97% accuracy".** It's measured against its own cluster labels, so rename it "assignment reproducibility". Validate externally by hand-labeling about 200 statutes (two raters) and reporting ARI/NMI or purity. Describe Domain 00 as a residual catch-all cluster (48%), which also explains why the Price Act and DRRM items land there. Rename the

benchmark “domains” to “premise groups”, or label them with the topic model’s assignments.

Smaller things

- Cut the duplicated paragraphs in 2.8.4/2.8.6.
- Replace each “This table is included to…” paragraph with one reading-guide sentence.
- Soften “proves”, “decisively” and “extraordinary”.
- Fix `¡` → `<` and the Table 4.20 caption (use `\textless` or math mode).
- Run a BibTeX audit script for duplicate keys (24/25, 89/90, 205/206). For [175], [223], [240], [1] and [33], either find the source that supports the claim or delete the claim, and drop the “70%” figure unless you find its source.

Reframing for abstract, conclusion and defense

State the contribution as three things: a retrieve-then-entail pipeline, a first controlled Philippine local-ordinance conflict benchmark, and a feasibility demonstration on LGU-class hardware. A sample abstract sentence: “On a controlled, adviser-verified 350-pair benchmark, fine-tuned DeBERTa-v3 reached 88.3% ± 3.4 grouped-CV accuracy; open-corpus full-document testing (Tier 2) remains limited by retrieval, motivating future work.”

Prepare answers to the likely panel questions:

- **Is it leaking?** Grouped splits, deduped data and masked-cue results answer this.
- **Why no human evaluation?** Institutional routing delays, with the protocol ready and a pilot subsample done.
- **Why hybrid if dense wins on MRR?** Stage 2 consumes top-50 and hybrid protects numeric/citation queries.
- **Why not an LLM?** Offline, privacy, hardware and hallucination constraints.

I can draft the revised abstract, limitations section or a script for the numbers registry and the leakage checks. Tell me which you want first.

Create a doc